

Research Paper 15 Summary

Snorkel: Rapid Training Data Creation with Weak Supervision

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Ratner et al. (2017), *VLDB Endowment*

1 Basic Information

- **Paper Title:** Snorkel: Rapid Training Data Creation with Weak Supervision
- **Authors:** Alexander Ratner, Stephen H. Bach, Henry Ehrenberg, et al. (Stanford University, 2017)
- **Core Contribution:** Solved the biggest bottleneck in Machine Learning by creating **Weak Supervision**, allowing AI training data to be generated programmatically without manual human labeling.

2 What is this Paper About? (In Simple Words)

To train an AI model to recognize industrial fires, you normally need millions of historical satellite points that have been manually labeled by humans. However, manually checking and labeling millions of satellite points across India is impossible.

- **The Snorkel Solution (Labeling Functions):** Instead of labeling points by hand, domain experts write small computer programs called **Labeling Functions (LFs)**. Each function applies a specific heuristic rule, spatial check, or physical formula.
- **The Generative Model:** Different rules may sometimes disagree or overlap. Snorkel's mathematical model evaluates the reliability and correlations of all labeling functions without needing ground-truth answers.
- **The Output:** It outputs clean, high-confidence probabilistic labels for millions of unlabelled data rows in minutes, which are then used to train our XGBoost model.

3 Key Concept: Writing Labeling Functions for Satellite Data

Instead of manual tagging, our pipeline writes programmatic rules:

- **Rule 1 (Spatial Proximity):** *If distance to nearest OSM Refinery < 300 m → Vote "Industrial".*
- **Rule 2 (Temporal Persistence):** *If heat observed ≥ 15 days in past month → Vote "Industrial".*
- **Rule 3 (Crop Season):** *If LandCover is Cropland AND Month is October AND Duration ≤ 3 days → Vote "Agricultural Burn".*

- **Rule 4 (Forest Context):** *If LandCover is Forest AND NDVI > 0.6 AND Persistence ≤ 5 days $\rightarrow$ Vote “Wildfire”.*
- **Rule 5 (Radiometric Heat):** *If Flame Temperature > 1300 K and Area < 50 m² $\rightarrow$ Vote “Industrial Flare”.*

4 How This Helps Our Project (PS26162)

4.1 1. Theoretical Defense for Training Data Generation

When evaluators or judges ask: “Where did you get labeled ground-truth data to train your AI model?”, citing Snorkel provides the formal scientific explanation:

“We used Stanford’s Snorkel weak supervision framework to combine OSM spatial buffers, temporal persistence, and physical temperature rules into programmatic labeling functions.”

4.2 2. Generating Massive Training Datasets in Minutes

Using Snorkel, our team can take 5+ years of raw, unlabeled NASA FIRMS satellite detections for India (millions of rows) and generate a rich, multi-class labeled training dataset automatically in a few minutes.

4.3 3. Resolving Conflicting and Ambiguous Rules

If an industrial factory is surrounded by farmland, a simple rule might be confused whether an alert is crop burning or factory heat. Snorkel’s generative model mathematically resolves these overlaps to assign the most accurate probability.